

Pioneer MainSearch Feature List — Drop Candidates

Branch: feature/mbr-batch-f | Date: 2026-05-18 | Based on overnight sweep (Olsen Astral / Exploris / MTAC, 6 files each).

Legend:  keep |  DROP (the 9 neighborhood features from combo_100lr20_drop_neighborhood)

Group 1 — Core PSM quality (14, all keep)

Feature	Status
fitted_manhattan_distance	
fitted_hellinger	
gof	
poisson	
err_norm	
weight	
total_ions	
y_count	
longest_y	
log2_intensity_explained	
Mox	
missed_cleavage	
sequence_length	
spectrum_peak_count	

Group 2 — Cross-precursor at-scan (2, keep)

Feature	Status
weight_ratio_at_scan	
weight_rank_at_scan	

Group 3 — Neighborhood (windowed PSM-quality) — DROP ALL 9

These came from the add_neighborhood_features! pass and were the entire deliverable of that step. Net-harmful on all three datasets.

Feature	Status
best_max_residual_3scan	
best_gof_5scan	
best_manhattan_5scan	
best_max_residual_5scan	

irt_dist_best_gof_5scan	✗
irt_dist_best_manhattan_5scan	✗
irt_dist_best_max_residual_5scan	✗
worst_max_residual_11scan	✗
worst_manhattan_11scan	✗

Group 4 — MS1 (6, all keep — Exploris depends on them)

Feature	Status
ms1_m0_intensity	✓
ms1_m1_intensity	✓
ms1_m0_mass_err_ppm	✓
ms1_m1_to_m0_ratio	✓
ms1_m1_to_m0_pred	✓
ms1_weight_apex_to_m0_apex_irt	✓

Group 5 — Per-rank M0 fragment intensities (4, keep)

Feature	Status
frag1_int	✓
frag2_int	✓
frag3_int	✓
frag4_int	✓

Group 6 — Fragment chromatogram (3, keep)

Feature	Status
frag_apex_dispersion_irt	✓
frag_corr_best_m0	✓
n_correlated_fragments	✓

Group 7 — RT / iRT / identity (5, keep)

Feature	Status
irt_diff	✓
irt_error	✓
irt_fwhm	✓
prec_mz	✓
n_scans	✓

Group 8 — Smoothness / Batch-E (4, keep)

Feature	Status
smoothness	✓
delta_frame_peak_center	✓
log_by_ratio_m0	✓
top3_ms2_mass_error_mean	✓

Group 9 — MBR features (ADVANCED_FEATURE_SET only — 6, keep)

These are in ADVANCED_FEATURE_SET (Pass-1 LGBM input) but **not** in PRESCORE_FEATURES (per-file LGBM).

Feature	Status
MBR_max_pair_prob_true	✓
MBR_log2_weight_ratio_true	✓
MBR_log2_explained_ratio_true	✓
MBR_best_irt_diff_true	✓
MBR_best_rt_diff_true	✓
MBR_is_missing_true	✓

Summary

List	Current count	After dropping 9	Δ
PRESCORE_FEATURES (per-file LGBM)	47	38	−9
ADVANCED_FEATURE_SET (Pass-1 LGBM)	52	43	−9

The 9 features to drop are identical between the two lists.

Follow-up worth measuring: with all 9 neighborhood features gone, the add_neighborhood_features! call itself becomes dead work (~4s per Astral file × 6 = ~24s further wall savings on Astral, on top of the −60s already measured). The cost of producing the columns is paid even though no LGBM consumes them. Quick A/B to confirm and skip the call.

Source: overnight LGBM HP × feature ablation sweep, Olsen Astral / Exploris / MTAC (6 files each), 2026-05-18. Full report: investigation_notes/overnight_lgbm_hp_feature_sweep_2026_05_18.md .